

ADR-005 : Migration from the Vanilla JS (Babel) interface to Vue.js

Contexte

Until now, a part of our user interface was developed in pure JavaScript (Vanilla JS) and compiled with Babel to ensure cross-browser compatibility. With the addition of new interactive features, direct DOM manipulation is becoming complex, difficult to maintain, and a source of bugs. To gain productivity and scalability, it is necessary to adopt a front-end architecture based on reusable components.

Options considérées

- Maintain the existing (Vanilla JS + Babel): Avoids an immediate overhaul, but technical debt and the difficulty of scaling will continue to grow.
- Migrate to React: A gigantic ecosystem and high market demand, but a potentially steep learning curve and concepts (like hooks or JSX) that could divide the team.
- Migrate to Vue.js: A progressive framework, a smoother learning curve with a clear separation (HTML template, script, style), and excellent native reactivity.

Décision

Adopt Vue.js for the interface redesign

Justification : Vue.js offers an excellent compromise between power and simplicity. Its progressive nature allows for smooth integration. Furthermore, using Vue.js modernizes our build chain (generally via Vite), which makes the Babel configuration obsolete and lightens our tooling. The team will be able to create complex interfaces more easily thanks to the component system and native state management.

Conséquences

- Positive : Drastic improvement in developer experience (DX) and development speed thanks to components
- Positive : Simplification of build tooling (abandoning Babel in favor of more modern solutions included in the Vue ecosystem).
- Négative : Addition of framework weight (bundle size) during page load compared to pure JavaScript.